Zoo502

Donnan equilibrium 2

The phenomenon of unequal distribution of ions across the two sides of a differentially permeable membrane.

• Discovered by Frederick Donnan in 1911.

• Given the name Donnan equilibrium.

Definition

• “If diffusible solutes are separated by a membrane that is freely permeable to water and electrolytes but totally impermeable to one species of ion, the diffusible solutes become unequally distributed between the two compartments”.

Example

Take pure water in two compartments, separated by a membrane. Add KCl to one of them. KCl will ionize.

• Ions diffuse through the membrane until the conc. of K+ and Cl- becomes equal on both sides of the membrane.

• Now add a salt “KA” in I. It will produce K+ and A-.

• K+ is diffusible, but A- is not. So A- will remain confined to I.

• K+ and Cl- quickly redistribute to a new equilibrium.

• Presence of A- will cause an unequal distribution of ions.

• K+ will be more conc. in I having Awhile

Cl- will be less conc. in it.

Q Difference between epinephrine and nor epinephrine 2

Both are neurotransmitters, epinephrine is secreted by renal Medulla and norepinephrine by nervous system. Both are vasoconstrictor work during fight or flight. Checmically same except norepinephrine has additional methyle group.
Norepinephrine acts on alpha-1, alpha-2, and beta-1 recpetors...works to raise the heart rate and modulate blood pressure
Epinephrine acts on alpha-1, alpha-2, beta-1, AND beta-2..effective anti-histamine and treatment for shock

Homeostasis and example 3

Homeostasis

• Walter Cannon (1929)
• The tendency of an organism to maintain relative internal stability

Homeostasis—Definition

• To keep internal fluctuations in a narrow range.
• The ability to protect internal environment from the harms of fluctuations in external environment.
• Homeostasis does not mean to keep a fixed internal environment.

Homeostasis—Examples

• Homeostatic regulation of water
• Homeostatic regulation of temperature
• Regulation of pH
• Regulation of glucose concentration

- Regulation of osmotic pressure
- Regulation of oxygen level
- Regulation of ion conc.

Environmental Fluctuations

- Most habitats are quite hostile to animal cells

Examples

- Freshwater is more dilute
- Seawater is more salty
- Environmental temperature may be too hot or too cold

Need for Homeostatic Mechanisms

- Environmental fluctuations are disruptive to the functions of cells, tissues and organs
- So, physiological regulatory systems to maintain relatively stable conditions are a necessity.

Note on axon 5

Axon

- Long and thick process with constant diameter.
- Arises from cell body.
- Lengths vary from few mm to more than a meter.

Axon—Functions

• Specialized to conduct signals away from the cell body.
• Carry information for long distances with high fidelity and without loss.

Axon—Structural Components

• Axons contain a jelly-like semi-fluid substance Axoplasm.
• A plasma membrane called as Axolemma.
• Have mitochondria and ER but lack Nissl's granules.

Axon Terminals

• Branched terminal ends of axons.
• At each terminal are small extensions telodendria with enlarged ends terminal knobs.
• Terminal knobs produce neurotransmitters.
• Axon terminals allow transmission of signals to other neurons, glands or muscle fibers.

Axon Hillock

• At the junction of cell body and axon is a coneshaped region called

axon hillock.
• Here signals are generated that travel down the axon.

Myelin Sheath

• A fatty myelin sheath cover the axons.
• It is secreted by Schwann cells.
• Myelin sheath is not continuous.
• There are nonmyelinated points called Nodes of Ranvier.

Negative feedback mechanism and two examples 5

Negative Feedback Systems

• The most common form of regulation
• The end product works to stop or slow down the process

Negative Feedback Example-1

• Breakdown of sugar in the cells generates ATP.
• Excess ATP "feeds back" and inhibits an enzyme

near the beginning of pathway.
• ATP production stops.

Negative Feedback Example-2

• Control of blood sugar (glucose) by insulin.
• Receptors sense a rise in blood sugar.
• Control center (pancreas) secretes insulin.
• Insulin lowers blood sugar levels.
• As blood sugar level reaches homeostasis, pancreas stops releasing insulin.

Give e.g of acclimatization along with effect? 2

Acclimatization

• An adaptive change within an individual animal.
• Happens due to chronic exposure to new, naturally occurring environmental conditions.

Acclimatization Example

• An animal migrates to a high altitude area
• Faces low oxygen partial pressure

Effects

• Lung ventilation rate

increases initially
• Ventilation drops to normal in few days
• Animal has acclimatized to the high-altitude

Q# Mode of action of nuerotransmitters? 2

Mode of Action of Neurotransmitters

• Neurotransmitter binds to specific protein receptors—ligand-gated ion channels in the postsynaptic membrane.
• Result: channels open & allow ionic current to flow into the postsynaptic cell and change its Vm.
• An AP is initiated.

Q# Phase of action potential? 3

Phases of AP

1. Stimulation
2. Rising phase (Depolarization)
3. Peak phase
4. Falling phase (Repolarization)
5. Undershoot (afterhyperpolarization)

Q# Hodgkin cycle? 3

Hodgkin Cycle

• A stimulus causes local depolarization of the membrane.

• Na+ channels open in response to initial depolarization.
• This increases Na+ conductance that allows Na+ to enter the cell, which cause further depolarization of membrane and decrease in Vm.
• This relationship between Na+ conductance (gNa) and Vm represents a type of positive feedback system and is termed as the Hodgkin cycle.
• It results in triggering of an action potential.

.....induce hallucinations

Q# Effects of neuromodulators.3

Neuromodulators affect neighboring neurons and modify their behavior at once.

• Neurotransmitters involved in altering the cellular properties and efficacy of synaptic transmission of multiple postsynaptic neurons.

• They act through Gprotein linked receptors that use voltage-gated ion channels for ion movements.
• Generation of response is slow, but lasts longer.

Neuromodulators

Major neuromodulators in the CNS include:
• Dopamine
• Serotonin
• Acetylcholine
• Histamine
• Norepinephrine
• Endorphins
• Enkephalins
• Substance-P

Effects of Neuromodulation

• Development of complex behavioral patterns e.g. happiness, exploration, revenge, reward, greed.
• Processes of thinking, cognition, planning, learning and memory.
• Behavioral problems: mood swings, sleep disturbances, feelings of stress, anxiety, anger and depression.

Q# Acetylcholine agnostic 2

Acetylcholine Agonists

• Molecules that mimic the action of a neurotransmitter are called agonists.
• Acetylcholine agonists are:
• Carbachol
• Nicotine
• Muscarine
• They can activate cholinergic synapses.

Q#Structure-function relationships

Function is Based on Structure

• Structural design is matched to functional demands
• Such relationships arise through evolution and natural selection

Example

A frog's feeding process involves:
• Skeletal muscles of limbs for jumping on prey
• Smooth muscles of stomach for grinding food
• Cardiac muscles of heart for

distribution of food
All these structures are adapted for their functions

Structural Specialization for Functions

- Skeletal muscles—evolved and adapted for movement of bones
- Smooth muscles of digestive tract—specialized for grinding and mixing.
- Cardiac muscles—specialized to pump and circulate blood

Q# Adaptation, Acclimatization and Acclimation

Adaptation

- A slowly occurring evolutionary process.
- Involves thousands of generations.
- Physiology of species becomes well matched to the environment.
- Ensures survival of the species.
- Generally not reversible.

Acclimatization

- An adaptive change within an individual animal.
- Happens due to chronic exposure to new, naturally occurring environmental conditions.

Acclimation

• Changes are induced experimentally in the laboratory

Both of these are:

• Acquired characters
• Restricted to one or few members of species
• Not inheritable
• Have no evolutionary significance
• Reversible

Q# Principle of homeostasis

To keep internal fluctuations in a narrow range.

• The ability to protect internal environment from the harms of fluctuations in external environment.
• Homeostasis does not mean to keep a fixed internal environment.

Q# Feedback control systems

Definition

• Mechanism of self-regulation of Biological processes.

Principle

• Output or product itself regulates the process

Significance

• Maintenance of homeostasis

Mechanism

Feedback controls regulate particular variables e.g.,
- Temperature
- Salinity
- pH

This regulation requires:
- Continuous sampling of controlled variables
- Respective corrective actions.

Q#Conformity and Regulation

Conformers
- Animals that conform to external changes
- Are unable to maintain homeostasis for internal conditions

Types
- Osmoconformers
- Oxyconformers
- Thermoconformers

Regulators
- Use internal control mechanisms to regulate internal conditions
- Maintain homeostasis

Types
- Osmoregulators
- Oxyregulators
- Thermoregulators